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N + - Network Fundamentals & Building Networks.

Introduction to OSPF.

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TASK 1.

❖ Open Cisco Packet Tracer and create a simple network with three routers connected in a triangle topology using serial or Ethernet links.

- **Creating a Three Router Triangle Topology:**

  Network Setup: Three routers (Router1, Router2, and Router3) were connected in a triangle topology using serial or Ethernet links.

  Connections:

  - Router1 ↔ Router2
  - Router2 ↔ Router3
  - Router3 ↔ Router1

  Result: A triangle network topology was created in Cisco Packet Tracer, providing multiple paths between the routers.

TASK 2.

❖ Assign IP addresses to all router interfaces in your Packet Tracer network and verify connectivity using the ping command.

- **IP Address Configuration and Connectivity Test:**

IP Address Configuration:

Router Interface	IP Address	Subnet Mask
R1 – S0/0/0	10.0.0.1	255.255.255.252
R2 – S0/0/0	10.0.0.2	255.255.255.252
R2 – S0/0/1	10.0.0.5	255.255.255.252
R3 – S0/0/0	10.0.0.6	255.255.255.252
R3 – S0/1/0	10.0.0.9	255.255.255.252
R1 – S0/0/1	10.0.0.10	255.255.255.252

Connectivity Test: The ping command was used to test connectivity between router interfaces.

Router# ping 10.0.0.2

Result: IP addresses were assigned to all router interfaces, and connectivity was verified using ping.

TASK 3.

❖ Configure OSPF on each router using the router ospf 1 command and the network command to advertise all directly connected networks.

- **Configure OSPF on All Routers:**
  The following commands were configured on each router to enable OSPF and advertise directly connected networks.

Router 1:

```
Router1(config)# router ospf 1
Router1(config-router)# network 10.0.0.0 0.0.0.3 area 0
Router1(config-router)# network 10.0.0.8 0.0.0.3 area 0
```

Router 2:

```
Router2(config)# router ospf 1
Router2(config-router)# network 10.0.0.0 0.0.0.3 area 0
Router2(config-router)# network 10.0.0.4 0.0.0.3 area 0
```

Router 3:

```
Router3(config)# router ospf 1
Router3(config-router)# network 10.0.0.4 0.0.0.3 area 0
Router3(config-router)# network 10.0.0.8 0.0.0.3 area 0
```

Result: OSPF was enabled on all three routers, and their connected networks were advertised in Area 0.

TASK 4.

❖ Observe and document the OSPF adjacency formation process by checking the output of the show ip ospf neighbor command on each router, Look for the FULL/DR or FULL/BDR state to confirm adjacency.

- **Observing OSPF Adjacency Formation:**

  Command used on each router:

  show ip ospf neighbor

  Observation: The command displays neighboring routers and their OSPF adjacency states. The FULL/DR or FULL/BDR state confirms that OSPF adjacency has been successfully established.

  Result: OSPF neighbor relationships were checked on all three routers to verify adjacency formation.

TASK 5.

❖ Change the OSPF cost on one interface using the ip ospf cost command and verify how it affects the OSPF routing table, Use show ip route ospf to see route changes after modifying the cost.

- **Changing OSPF Cost and Verifying Routes:**

Change OSPF Cost: The interface cost was changed to 100 on the selected router interface.

Router(config)# interface serial0/0/0
Router(config-if)# ip ospf cost 100

Verify Routing Table:

Router# show ip route ospf

Observation: OSPF recalculates the best path based on the updated cost. A higher cost path may cause traffic to use an alternative route>

Result: The OSPF routing table was checked to observe route changes after modifying the interface cost.